MILITARY SPECIFICATION
LOGIC FLIP-FLOP: 10526505

1. SCOPE

1.1 This specification covers one type of logic flip-flop circuit board, part of the Gun Direction Computer: M18. The logic flip-flop is used to provide logic gating and storage of binary information for the computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18, Manufacturing processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component parts

DRAWINGS

U. S. Army Munitions Command

F10526505	Assembly, Logic Flip-Flop

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

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(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The logic flip-flop shall be manufactured in accordance with Drawing F10526505 and drawings pertaining thereto.

3.2 General specifications.- The logic flip-flop shall meet the following requirements of MIL-F-13926 and MIL-C-46357:

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The logic flip-flop shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the logic flip-flop for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the logic flip-flop shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The logic flip-flop shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3. Temperatures.

3.3.3.1 Operating.- The logic flip-flop shall meet the requirements of 3.4.3 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The logic flip-flop shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F

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3.3.4 Humidity.

3.3.4.1 Operating.- The logic flip-flop shall meet the requirements of 3.4.3 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F.

3.3.4.2 Non-operating.- The logic flip-flop shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The logic flip-flop shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The logic flip-flop shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine spray of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the logic flip-flop shall meet the performance requirements of this specification at standard ambient temperatures between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%.

3.4.1 Logic flip-flop (true state).- With the logic flip-flop in the true state, the following conditions shall exist when the output'(prime) is loaded and the output is unloaded:

a. The saturation voltages of the "on" transistors shall not exceed 0.3 volts (v).

b. The voltage at the base of the "off" input transistor shall be greater than 0.18 v with respect to the +1.25v supply.

c. The leakage current through the output transistor shall not exceed 15 microamps (ua).

3.4.2 Logic flip-flop (false state).- With the logic flip-flop in the false state, the following conditions shall exist when the output' is unloaded and the output is loaded:

a. The saturation voltages of the "on" transistors shall not exceed 0.3v.

b. The voltage at the base of the "off" input transistor shall be greater than 0.18v with respect to the +1.25v supply.

c. The leakage current through the output' transistor shall not exceed 15ua.

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3.4.3 Logic flip-flop (full load dynamic conditions).

3.4.3.1 Minimum clock pulse.- When a self-gated flip-flop (see Figure 1) is loaded on both sides and triggered by a 3.8 ± 0.3v clock signal, the output and output' signals shall be -0.15 ± .15v to -6 ± 1v square waves. The flip-flop shall not toggle, when the clock amplitude is reduced to 1.5 ± 0.1v.

3.4.4 Diodes.

3.4.4.1 Logic diode, 10525848 -(type IN99) voltage drop.- The forward voltage drop across a standard logic diode shall be less than 0.6v at an operating current of 1.5ma.

3.4.4.2 Logic diode (10525848)-reverse current.- The reverse current through the standard logic diodes shall not exceed 20μa at 15v.

3.4.4.3 "And" diodes (10525849)-voltage drop.- The forward voltage drop across the primary "and" diodes shall be less than 0.5v at 1.5 milliamps (ma).

3.4.4.4 "And" diodes (10525849)-reverse current.- The reverse current through the primary "and" diodes shall not exceed 10μa at 15v.

3.4.4.5 Test diodes (10525855)-voltage drop.- The forward voltage drop across the test diodes shall be less than 0.7v at 0.3ma.

3.4.4.6 Test diodes (10525855)-reverse current.- The reverse current through the test diodes shall not exceed 0.2μa at 15v.

4. QUALITY ASSURANCE PROVISIONS

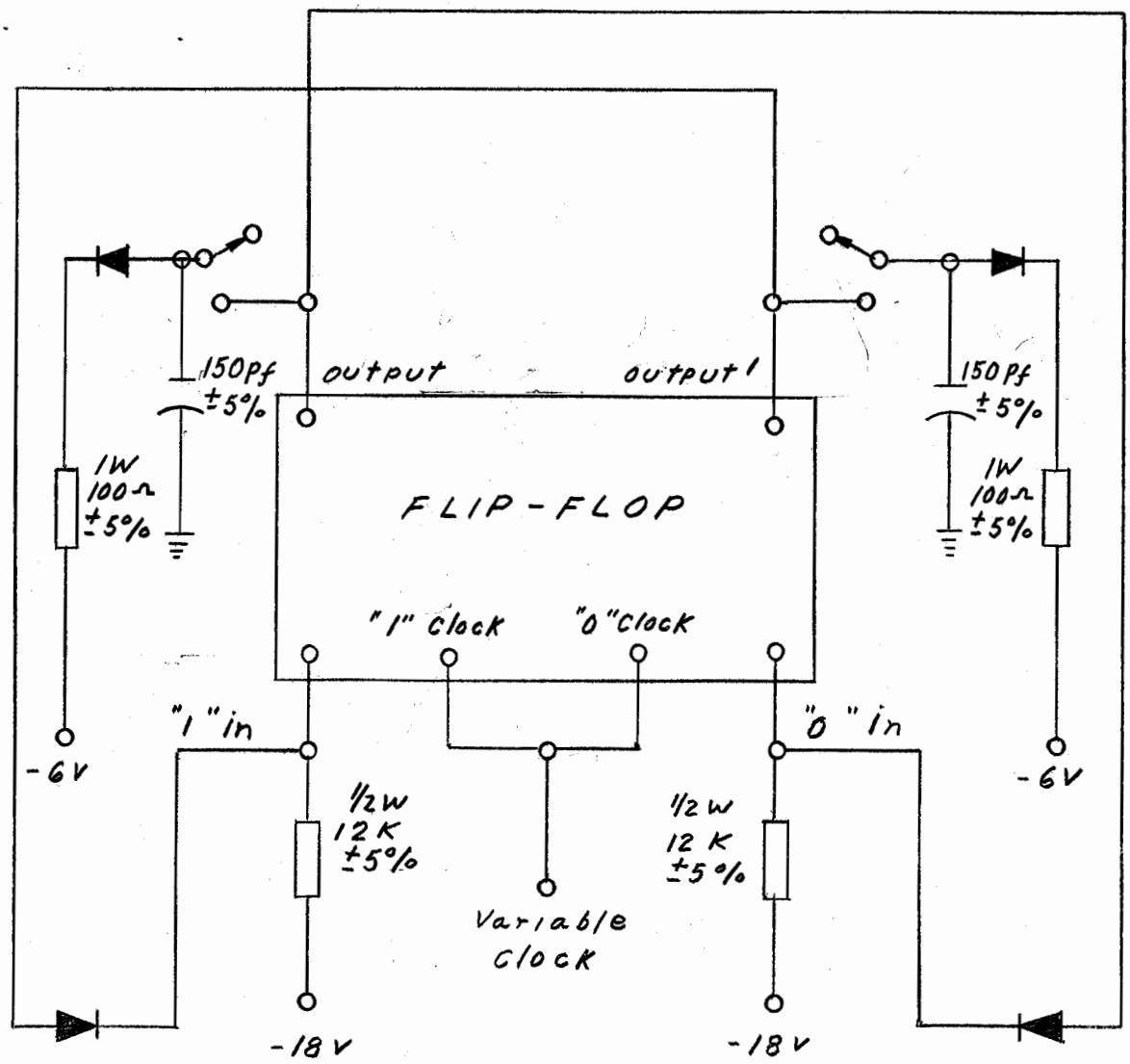
4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

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NOTE: Diodes are 105 2584 8

FIGURE 1 - FLIP-FLOP SELF-GATING SYSTEM

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4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 percent inspection)

	Characteristics	Requirement	Test Procedures
101.	Logic flip-flop (true state	3.4.1	4.5.2.1.1
102.	Logic flip-flop (false state)	3.4.2	4.5.2.1.2
103.	Minimum clock pulse	3.4.3.1	4.5.2.1.3.1
104.	Logic diodes-voltage drop	3.4.4.1	4.5.2.1.4
105.	Logic diodes-reverse current	3.4.4.2	4.5.2.1.5
106.	"And" diodes-voltage drop	3.4.4.3	4.5.2.1.6
107.	"And" diodes-reverse current	3.4.4.4	4.5.2.1.7
108.	Test diodes-voltage drop and reverse current	3.4.4.5, 3.4.4.6	4.5.2.1.8

4.3 Standard sampling.

4.3.1 General.- One logic flip-flop shall be selected at random by a Government representative as a standard sample from each 100 produced. The standard sample shall meet the requirements and tests in Table II.

TABLE II

	Characteristics	Requirements	Test Procedures
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

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4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

	Characteristics	Requirements	Test Procedures
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity-operating	3.3.4.1, 3.4.3	4.5.1.3.2, 3.5.2.1.3
306.	Humidity-non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1, 3.4.3, 3.4.3.1	4.5.1.5, 4.5.2.1.3, 4.5.3.1
309.	Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

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Item Number	Description
1. Power Supply	Plus 15 volts ± 2 percent, current rating greater than or equal to 2 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
2. Power Supply	Minus 18 volts ± 2 percent, current rating greater than or equal to 100 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from D.C. to 500 kc.
3. Power Supply	Minus 6 volts ± 2 percent, current rating greater than or equal to 250 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
4. Power Supply	Plus 1.25 volts ± 2 percent, current rating greater than or equal to 50 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
5. Power Supply	Plus 6 volts ± 2 percent, current rating greater than or equal to 10 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
6. Power Supply	Plus 35 ± 2 volts, current rating greater than or equal to 3.0 milliamperes, peak-to-peak ripple less than 100 millivolts, output impedance less than 5 ohms from D.C. to 20kc.
7. Power Supply	Minus 25 ± 2 volts, current rating greater than or equal to 2 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 5 ohms from D.C. to 20kc, connected through a 10k ohm ± 5 percent ½watt resistor. (See Figure 2).
8. Clock Source	Variable amplitude to supply the 2 following waveforms: (a) Minus 3.8 ± 0.3 volts for 1.8 ± 0.1 microseconds (true) Minus 0.15 ± 0.15 volts for 0.5 ± 0.05 microseconds (false). (b) Minus 1.5 ± 0.1 volts for 1.8 ± 0.1 microseconds (true) Minus 0.15 ± 0.15 volts for 0.5 ± 0.05 microseconds (false). Clock rise time (10 to 90 percent amplitude) shall be less than 150 millimicroseconds. Rise time is measured during the true to false transition. Clock fall time (10 to 90 percent amplitude) shall be 0.4 microseconds or less.

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9.	Oscilloscope	Dual channel rise time of scope plus measuring probe 50 millimicroseconds or less, sweep period 50 millicroseconds per centimeter or greater, input resistance greater than 8 megohms , imput capacitance less than 25 micromicrofarad.
10.	Voltmeter	D.C., 0-50 volt range; input impedance greater than 10 megohms.
11.	Resistor	56K ohm ± 5%, 1/2 w
12.	Resistor (4)	12K ohm ± 1%, 1/2 w
13.	Resistor	1.3K ohm ± 5%, 1/2 w
14.	Resistors (2)	100 ohm ± 5%, 1w
15.	Capacitor (2)	150 pf ± 5%



NOTE: Apply voltages only momentarily

FIGURE 2 - Set and Reset Voltages - Flip-flop

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the logic flip-flop rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the logic flip-flop shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that received the logic flip-flop in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical

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plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test both circuits of the unit shall be operated in accordance with 4.5.2.1.3 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The logic flip-flop shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.1 and 4.5.2.1.3.1 for compliance with 3.3.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the logic flip-flop shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.1.3 to determine compliance with the requirements of 3.4.3 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.1 to 4.5.2.1.3.1 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.- The 2 logic flip-flop circuits (Circuit 1 and Circuit 2) on the module shall each be subjected to the tests specified in 4.5.2.1 to 4.5.2.1.3.1 inclusive and shall meet the requirements referenced therein when the corresponding inputs, outputs, pins and test points as shown in Table IV are respectively compared.

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TABLE IV
Corresponding Points

Flip-Flop (Circuit 1)	Flip-Flop (Circuit 2)
Output' - pin 27B	Output' - 36B
Output - pin 28B	Output - 3ϐB
TP7	TP8
TP9	TP10
Pin 39B	Pin 39B
Pin 13B	Pin 13B
Pin 31B	Pin 31B
Pin 22B	Pin 22B
TP11	TP12
TP13	TP14
Pin 38B	Pin 38B

4.5.2.1 Logic flip-flop, circuit I.- Connect the power supplies to the respective pins as specified in Table V. The D.C. return of each power supply specified in Table V shall be connected to pin 31B, and the conditions for flip-flop, circuit 1 of Table IV shall apply.

TABLE V - Power Supply Connections

Pin	Power Supply
12B	+6V
22B	-6V
39B	+1.25V

4.5.2.1.1 Logic flip-flop (true state).- Apply the load shown in Figure I to the output' (pin 27B). Do not load the output (pin 28B). Momentarily apply the set voltage specified in Figure 2 to TP7. The voltages as measured below shall be within the specified tolerances or limits to determine conformance with the requirements of 3.4.1:

(a) The voltage from TP9 to pin 39B shall be minus 0.15 ± 0.15 volts.
(b) The voltage from pin 27B to pin 31B shall be minus 0.15 ± 0.15 volts.
(c) The voltage from TP7 to pin 39B shall be plus 0.18 volts or greater.
(d) The voltage from pin 28B to pin 22B shall be less than 15 millivolts.

4.5.2.1.2 Logic flip-flop (false state).- Apply the load shown in Figure I to the output (pin 28B). Do not load the output' (pin 27B). Momentarily apply the reset voltage specified in Figure 2 to TP13. The voltages as measured below shall be within the specified tolerances or limits to determine conformance with the requirements of 3.4.2:

(a) The voltage from TP11 to pin 39B shall be minus 0.15 ± 0.15 volts.
(b) The voltage from pin 28B to pin 31B shall be minus 0.15 ± 0.15 volts.
(c) The voltage from TP13 to pin 39B shall be plus 0.18 volts or greater.
(d) The voltage from pin 27B to pin 22B shall be less than 15 millivolts.

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4.5.2.1.3 Logic flip-flop (full load dynamic conditions).- The following test shall be conducted with the pins and power supplies connected as specified in 4.5.2.1 Loads shall be applied to both the output and the output' (see Figure 1) when testing for conformance to the requirements under 3.4.3.

4.5.2.1.3.1 Minimum clock pulse.- Make the following connections and measurements to determine conformance with the requirements of 3.4.3.1:

(a) Apply the 3.8 volt clock signal (Item 8A of 4.4.1.1) to pins 38B and 13B.

(b) The voltage level at the output shall be monitored. It shall be a square wave alternating between the levels of minus 0.15 ± 0.15 volts and minus 6 ± 1 volt, and shall make a transition every time a clock pulse occurs.

(c) The voltage level at the output' shall be monitored. It shall be a square wave alternating between the levels of minus 0.15 ± 0.15 volts and minus 6 ± 1 volt, and shall make a transition every time a clock pulse occurs.

(d) Remove the 3.8 volt clock signal and apply the 1.5 volt clock signal (Item 8B of 4.4.1.1) to pins 38B and 13B.

(e) The output and output' signals shall be monitored. The flip-flop shall not trigger.

4.5.2.1.4 Logic diodes (10525848) voltage drop.- All external connections shall be removed from the module including power supplies and power ground. Compliance with 3.4.4.1 shall be determined by connecting pins 55A, 55B, 54B and 54A to pin 24A. Connect the minus 18 volt power supply to pin 39A. Connect the pin specified in Table VI, Column A to ground and measure the voltage level from ground to the pin specified in Table VI, Column B with the pin specified in Table VI, Column B connected to the plus 6 volt power supply through a 56K ± 5%, 1/2 W resistor. The measured voltage shall be more positive than -.40 volts.

TABLE VI

A	B
Pin 53B	Pin 53A
Pin 53A	Pin 53B
Pin 14B	Pin 16B
Pin 16B	Pin 14B
Pin 52B	Pin 50B
Pin 51B	Pin 50B
Pin 50B	Pin 51B
Pin 57B	Pin 56B
Pin 56A	Pin 56B
Pin 56B	Pin 56A
Pin 57A	Pin 58A
Pin 58B	Pin 58A
Pin 58A	Pin 58B
Pin 2B	Pin 1A
Pin 3A	Pin 1A
Pin 1B	Pin 1A
Pin 2A	Pin 1A

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TABLE VI (Continued)

A	B
Pin 1A	Pin 2A
Pin 45A	Pin 45B
Pin 46A	Pin 45B
Pin 42A	Pin 45B
Pin 40A	Pin 45B
Pin 45B	Pin 40A
Pin 60A	Pin 59B
Pin 59A	Pin 59B
Pin 59B	Pin 59A
Pin 6A	Pin 9B
Pin 7B	Pin 9B
Pin 7A	Pin 9B
Pin 8B	Pin 9B
Pin 8A	Pin 9B
Pin 9B	Pin 8A
Pin 42B	Pin 43A
Pin 43B	Pin 43A
Pin 44B	Pin 43A
Pin 44A	Pin 43A
Pin 43A	Pin 44A
Pin 34A	Pin 38A
Pin 35A	Pin 38A
Pin 36A	Pin 38A
Pin 37A	Pin 38A
Pin 38A	Pin 37A
Pin 31A	Pin 35B
Pin 32A	Pin 35B
Pin 33A	Pin 35B
Pin 35B	Pin 33A
Pin 65A	Pin 64B
Pin 65B	Pin 64B
Pin 64A	Pin 64B
Pin 64B	Pin 64A
Pin 30A	Pin 28A
Pin 30B	Pin 28A
Pin 29A	Pin 28A
Pin 28A	Pin 29A
Pin 12A	Pin 14A
Pin 15B	Pin 14A
Pin 13A	Pin 14A
Pin 14A	Pin 13A
Pin 17A	Pin 15A
Pin 16A	Pin 15A
Pin 18A	Pin 15A
Pin 15A	Pin 18A
Pin 9A	Pin 10B
Pin 10B	Pin 9A
Pin 11A	Pin 9A

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TABLE VI (Continued)

A	B
Pin 11A	Pin 10A
Pin 11B	Pin 10A
Pin 10A	Pin 11B
Pin 61B	Pin 62B
Pin 62B	Pin 61B
Pin 62A	Pin 61B
Pin 60B	Pin 61B
Pin 60B	Pin 61A
Pin 63B	Pin 61A
Pin 63A	Pin 61A
Pin 61A	Pin 63A
Pin 5B	Pin 6B
Pin 6B	Pin 5B
Pin 4B	Pin 5B
Pin 5A	Pin 5B
Pin 5A	Pin 4A
Pin 3B	Pin 4A
Pin 4A	Pin 3B

4.5.2.1.4.1 Connect the pin specified in Table VII, Column A to ground and measure the voltage level from ground to the pin specified in Table VII, Column B. The measured voltage shall be more positive than -.60 volts.

TABLE VII

A	B
Pin 22B	Pin 25B
Pin 22B	Pin 19A
Pin 22B	Pin 19B
Pin 22B	Pin 24A

4.5.2.1.5 Logic diode (10525848) reverse current.- Disconnect the minus 18 volt power supply from pin 39A and pin 24B. Connect the plus 15 volt power supply to pin 39A. Make the following connections and measurements to determine conformance with 3.4.4.2.

4.5.2.1.5.1 Connect the pins specified in Table VI, Column A to ground through a 12K ohm resistor and measure the voltage level from ground to the pin specified in Column A. The measured voltage shall be less than ± .180 volts for all pins except pin 5A, pin 11A, and pin 60B. For pin 5A, pin 11A, and pin 60B, the measured value shall be less than ± .360 volts.

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4.5.2.1.5.2 Connect the pin specified in Table VII, Column A to ground and measure the voltage level from pin 39A to the pin specified in Table VII, Column B. The measured voltage difference shall be less than + .180 volts.

4.5.2.1.6 "And" diodes (10525849) voltage drop.- Connect pins 55A, 55B, 54B and 54A to pin 24A. Connect the minus 18 volt power supply to pin 39A and pin 24B. Connect the pin specified in Table VIII, column A to ground and measure the voltage level from ground to the pin specified in Table VIII, column B with the pin specified in Table VIII, column B connected to the minus 18 volt power supply through a 1.3K ± 5%, 1/2W resistor. The measured voltage shall be more positive than -.50 volts to determine conformance with 3.4.4.3.

TABLE VIII

A	B
Pin 29B	Pin 25B
Pin 18B	Pin 25B
Pin 21A	Pin 19A
Pin 20A	Pin 19A
Pin 23A	Pin 19B
Pin 23B	Pin 19B
Pin 22A	Pin 19B
Pin 27A	Pin 24A
Pin 26A	Pin 24A
Pin 25A	Pin 24A
Pin 50A	Pin 24A
Pin 47B	Pin 24A
Pin 52A	Pin 24A
Pin 51A	Pin 24A
Pin 48A	Pin 24A
Pin 49B	Pin 24A
Pin 49A	Pin 24A
Pin 47A	Pin 24A
Pin 48B	Pin 24A

4.5.2.1.7 "And" diodes (10525849) reverse current.- Disconnect the minus 18 volt power supply from pin 39A and pin 24B. Connect the +15 volt power supply to pin 39A. Connect the pin specified in Table VIII, Column A to ground and measure the voltage level from pin 39A to the pin specified in Table VIII, column B. The measured voltage difference shall be less than .180 volts to determine conformance with 3.4.4.4.

4.5.2.1.8 Miscellaneous diode tests - voltage drop and reverse current.- Make the following connections and measurements to determine conformance with the requirements of 3.4.4.1, 3.4.4.2, 3.4.4.5, and 3.4.4.6:

4.5.2.1.8.1 Remove the plus 15 volt power supply from pin 39A and make the following connections:

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(a) Connect TP21 to the minus 18 volt power supply through a 12K ± 1%, 1/2W resistor. Ground pin 21B. The voltage from pin 21B to TP21 shall be more positive than -.60 volts. Remove the ground from pin 21B and ground pin 34B. The voltage from pin 34B to TP21 shall be more positive than -.60 volts.

(b) Remove the ground from pin 34B and remove the minus 18 volt power supply from TP21. Connect TP22 to the minus 18 volt power supply through a 12K ± 1%, 1/2W resistor. Ground pin 37B. The voltage from pin 37B to TP22 shall be more positive than -.60 volts. Remove the ground from pin 37B and ground pin 34B. The voltage from pin 34B to TP22 shall be more positive than -.60 volts.

(c) Remove the ground from pin 34B and remove the minus 18 volt power supply from TP22. Connect the minus 18 volt power supply to pin 24B. Ground TP21. The voltage from TP21 to pin 20B and to pin 17B shall be more positive than -.60 volts. Remove the ground from TP21. Ground TP19. The voltage from TP19 to pin 17B shall be more positive than -.60 volts. Remove the ground from TP19. Ground TP1. The voltage from TP1 to pin 20B shall be more positive than -.60 volts.

(d) Remove the ground from TP1. Ground TP22. The voltage from TP22 to pin 40B and to pin 46B shall be more positive than -.60 volts. Remove the ground from TP22. Ground TP20. The voltage from TP20 to pin 46B shall be more positive than -.60 volts. Remove the ground from TP20. Ground TP2. The voltage from TP2 to pin 40B shall be more positive than -.60 volts.

4.5.2.1.8.2 Remove the ground from TP2 and remove the minus 18 volt power supply from pin 24B. Connect TP21 to the plus 15 volt power supply through a 12K ± 1%, 1/2W resistor. Ground pin 21B. The voltage between the plus 15 volt power supply and TP21 shall be less than .180 volt. Remove the ground from pin 21B and ground pin 34B. The voltage between the plus 15 volt power supply and TP21 shall be less than .180 volt.

4.5.2.1.8.2.1 Remove the ground from pin 34B and remove the plus 15 volt power supply from TP21. Connect TP22 to the plus 15 volt power supply through a 12K ± 1%, 1/2W resistor. Ground pin 37B. The voltage between the plus 15 volt power supply and TP22 shall be less than .180 volts. Remove ground from pin 37B. Ground pin 34B. The voltage between the plus 15 volt power supply and TP22 shall be less than .180 volt.

4.5.2.1.8.3 Remove the ground from pin 34B and remove the plus 15 volt power supply from TP22. Connect the plus 15 volt power supply to pin 24B. Ground TP21. The voltage between pin 24B and pin 20B and between pin 24B and pin 17B shall be less than 1.0 volt. Remove the ground from TP21. Ground TP19. The voltage between pin 24B and pin 17B shall be less than 1.0 volt. Remove the ground from TP19. Ground TP1. The voltage between pin 24B and pin 20B shall be less than 1.0 volt.

4.5.2.1.8.3.1 Remove the ground from TP1. Ground TP22. The voltage between pin 24B and pin 46B, and between pin 24B and pin 40B shall be less than 1.0 volt. Remove the ground from TP22. Ground TP20. The voltage between pin 24B and pin 46B shall be less than 1.0 volt. Remove the ground from TP20. Ground TP2. The voltage between pin 24B and pin 40B shall be less than 1.0 volt.

4.5.2.1.8.4 Remove the plus 15 volt power supply from pin 24B and remove the ground from TP2. Connect the minus 18 volt power supply to pin 24B.

(a) Ground pin 32B. The voltage from ground to TP5 and TP6 shall not exceed .7 volts.

(b) Remove the ground from pin 32B. Connect pin 26B to ground. The voltage from ground to TP17 and to TP18 shall not exceed 0.7 volts.

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(c) Remove the ground from pin 26B. Connect TP21 to ground. The voltage from ground to TP5 and to TP17 shall not exceed 0.6 volts.

(d) Remove the ground from TP21. Connect TP22 to ground. The voltage from ground to TP6 and to TP18 shall not exceed 0.6 volts.

(e) Remove the ground from TP22. Connect pin 28B to ground. The voltage from ground to TP5 shall not exceed 0.6 volts.

(f) Remove the ground from pin 28B. Connect pin 27B to ground. The voltage from ground to TP17 shall not exceed 0.6 volts.

(g) Remove the ground from pin 27B. Connect pin 33B to ground. The voltage from ground to TP6 shall not exceed 0.6 volts.

(h) Remove the ground from pin 33B. Connect pin 36B to ground. The voltage from ground to TP18 shall not exceed 0.6 volts.

4.5.2.1.8.5 Remove the minus 18 volt power supply from pin 24B. Connect the plus 15 volt power supply to pin 24B. Remove the ground from pin 36B.

(a) Connect pin 32B to ground. The voltage from pin 24B to TP5 and to TP6 shall not exceed 14.4 millivolts.

(b) Remove the ground from pin 32B. Connect TP21 to ground. The voltage from pin 24B to TP5 and to TP17 shall not exceed 1.4 volts.

(c) Remove the ground from TP21. Connect pin 26B to ground. The voltage from pin 24B to TP17 and to TP18 shall not exceed 14.4 millivolts.

(d) Remove the ground from pin 26B. Connect TP22 to ground. The voltage from pin 24B to TP18 and to TP6 shall not exceed 1.4 volts.

(e) Remove the ground from TP22. Connect pin 28B to ground. The voltage from pin 24B to TP5 shall not exceed 1.4 volts.

(f) Remove the ground from pin 28B. Connect pin 27B to ground. The voltage from pin 24B to TP17 shall not exceed 1.4 volts.

(g) Remove the ground from pin 27B. Connect pin 33B to ground. The voltage from pin 24B to TP6 shall not exceed 1.4 volts.

(h) Remove the ground from pin 33B. Connect pin 36B to ground. The voltage from pin 24B to TP18 shall not exceed 1.4 volts.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Master Packaging Data Sheet M14002.

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6. NOTES

6.1 Intended use.- The logic flip-flop circuit board consists of 2 bistable multi-vibrator circuits used for performing logical computer processes such as counting and storing information. Each bistable multivibrator is capable of driving 28 standard logical gates. The circuit board also contains logic gates which are connected to various points in the computer to mechanize the logical processes which the computer performs.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title number and date of this specification.
(b) Selection of applicable level of preservation, packaging, and packing.
(c) Applicable stock numbers.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each logic flip-flop.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A189Y

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